

GURURAJ H A

Senior Hardware Component Engineer | Lifecycle Management | Design Engineering

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PROFESSIONAL SUMMARY

Results-driven Senior Hardware Component Engineer with 7+ years of expertise in hardware design, component lifecycle management, and obsolescence mitigation for automotive and industrial sectors. Proven track record of driving 30% reduction in obsolescence risk, 20% improvement in system reliability (MTBF), and 18% faster project delivery through strategic component engineering, circuit design, and cross-functional collaboration. Expert in analog/digital circuit design, PCB layout, schematic capture, hardware testing, and qualification using industry-standard tools (ORCAD, Altium Designer, Mentor Graphics). Deep proficiency in PLM/ERP systems (SAP, Teamcenter, Agile), component qualification protocols, and regulatory compliance (ISO 9001/26262, IATF 16949, RoHS, REACH). Demonstrated ability to design, debug, and validate hardware from concept through production while managing complex supply chain challenges during semiconductor shortages.

KEY ACHIEVEMENTS

- Drove **30% reduction in component obsolescence risk** through proactive lifecycle monitoring, alternate component qualification, and comprehensive Form-Fit-Function (FFF) validation—ensuring zero production interruptions during semiconductor crisis
- Improved system reliability (MTBF) by 20% while reducing component costs by 15% via strategic sourcing, derating analysis, and second-source qualification across 80+ BOMs
- Accelerated hardware **project delivery by 18%** through streamlined ECR/ECN/ECO processes, standardized component evaluation protocols, and enhanced PLM system automation
- Reduced **component verification cycle time by 25%** by developing automated validation methodologies and standardized qualification procedures aligned with IPC-A-620 standards
- Led design and development of automotive ECUs, instrument clusters, and HEV power electronics meeting ISO 26262 functional safety requirements
- Managed 150+ PCN/PDN annually with 100% on-time assessment, coordinating seamless component transitions and supplier negotiations across cross-functional teams
- Completed 6+ Kaizen projects improving component lead times by 15%, BOM accuracy by 20%, achieving operational excellence in 60+ PCBA risk mitigation initiatives

AWARDS & RECOGNITION

 **HCL Excellence Award**
Operational Excellence, 2024

 **Manufacturing Excellence**
Supporting Award – Yazaki

CORE COMPETENCIES

Hardware Design & Development: Analog/Digital Circuit Design | Schematic Capture (ORCAD, Altium Designer, Mentor Graphics PADS) | PCBA risk mitigation | Hardware Debugging | Board Bring-up | Prototyping | Design for Manufacturing (DFM) | Design for Test (DFT) | Design for Assembly (DFA) | Thermal Analysis | EMI/EMC Testing

Component Engineering: Component Selection & Qualification | EOL/Obsolescence Analysis | Alternate Component Evaluation | Form-Fit-Function (FFF) Validation | Second Source Qualification | Component Derating (MIL-STD) | AVL Management | Sample Library Administration | Part Number Creation | PCN/PDN Management

Testing & Validation: Hardware Qualification Testing | Environmental Stress Testing (Temperature, Vibration, Humidity) | Digital Oscilloscope (DSO) | Multimeter | Function Generator | DC Power Supply | EMI/EMC Test Equipment | Reliability Evaluation (HTOL, Temperature Cycling, Moisture Sensitivity)

Lifecycle & Change Management: Engineering Change Requests (ECR/ECO/ECN) | BOM Management | BOM Scrubbing | Production BOM Reviews | PLM Systems (SAP, Teamcenter, Agile, Windchill) | IHS Markit | Silicon Expert | Part Miner | Z2Data

Quality & Reliability: DFMEA | Root Cause Analysis (RCA) | Failure Analysis | MTBF/MTTF | 8D Problem Solving | 5WHY Analysis | A3 Problem Solving | CAPA | APQP | PPAP | Manufacturing Support

Communication Protocols: CAN | LIN | Ethernet | Ether CAT | USB 2.0/3.0 | RS-232 | RS-485 | I2C | SPI | UART

Compliance & Standards: ISO 9001/14001/26262/13485 | IATF 16949 | AS9100 | RoHS | REACH | CMRT | CE | UL | FCC | IPC/WHMA-A-620 | MIL-STD | Export Classification (ITAR, EAR, ECCN)

Process Excellence: APQP | PPAP | Kaizen | Six Sigma | Lean Manufacturing | VAVE | Cost Reduction Initiatives | Inventory Optimization | Supplier Qualification | Vendor Management | RFQ/RQP Management | Contract Manufacturer Support

PROFESSIONAL EXPERIENCE

TECHNICAL LEAD – COMPONENT ENGINEERING

Diya Instruments | Bengaluru, India | June 2025 - Present

Owns end-to-end system sustaining engineering for precision NDT instruments, driving hardware lifecycle governance, component risk management, and BOM integrity.

- ◆ **Hardware Lifecycle Management:** Directed end-to-end lifecycle management for precision NDT instruments—integrating sensors, display electronics, power management ICs, and communication interfaces into reliable, cost-effective assemblies.
- ◆ **Alternate Component Qualification:** Qualified 100% FFF alternates for critical sensors and display modules via AQL-based reliability plans, achieving zero production halts from component shortages.
- ◆ **VAVE & Dual-Sourcing:** Reduced material costs by 12% through dual-sourcing initiatives and technical alternate evaluations while maintaining full RoHS/REACH compliance.
- ◆ **DFM/DFT Standardization:** Reduced pilot build defects by 28% and improved manufacturing reliability by 20% through systematic DFM/DFT protocol standardization and supplier coordination.
- ◆ **Regulatory Compliance:** Executed global BOM scrubbing (RoHS, REACH, CMRT) ensuring multi-regional market readiness across all product lines.
- ◆ **Supplier Liaison:** Collaborated with global vendors on component specification validation, improving manufacturing reliability by 20%.

★ **RESULTS:** 28% reduction in pilot build defects · 20% improvement in manufacturing reliability · 12% material cost savings · Zero production stoppages

TECHNICAL LEAD – COMPONENT ENGINEERING

HCL Technologies | Bengaluru, India | March 2023 – June 2025

Led comprehensive hardware component engineering programs for mission-critical automotive and industrial electronics, driving lifecycle management, obsolescence mitigation, design support, and cross-functional collaboration across engineering, supply chain, quality, and manufacturing teams.

- ◆ **Lifecycle & Architecture Governance:** Led component engineering for mission-critical aerospace and industrial electronics programmes — managing 200+ components (MCUs, power ICs, semiconductors, passives, discrete) with 99.5% availability under AS9100, ISO 9001, and ISO 26262 standards.
- ◆ **PLM Automation & Digital Transformation:** Architected a Power BI Lifecycle Dashboard integrating SAP, IHS Markit, and supplier data in real-time — automating PCN/PDN alert workflows and eliminating 80% of manual review effort. Awarded HCL Excellence Award 2024.
- ◆ **Obsolescence Risk Mitigation:** Achieved 30% obsolescence risk reduction through proactive FFF validation (parametric analysis, electrical characterization, HTOL, temperature cycling) across 80+ BOMs.
- ◆ **PCN/PDN Governance:** Managed 150+ Product Change Notifications annually — 100% on-time impact assessments with zero unplanned production global semiconductor supply crisis.
- ◆ **Reliability Engineering:** Improved system MTBF by 20% via MIL-STD derating analysis, parametric stress calculations, and thermal simulations aligned with IPC-A-620 qualification protocols.
- ◆ **Change Management:** Streamlined engineering change processes for 60+ PCBA designs via Teamcenter PLM automation, reducing delivery time by 18% and improving stakeholder alignment.
- ◆ **PLM Data Governance:** Created and maintained 500+ part numbers in SAP, Teamcenter, and Agile PLM with 99.8% accuracy, supporting global procurement and manufacturing.
- ◆ **Cost Reduction (VAVE):** Drove 15% annual component cost reduction through supplier consolidation, VAVE workshops, and alternate qualification across 80+ BOMs.

- ◆ Created and maintained 500+ part numbers in PLM systems (SAP, Teamcenter, Agile) with 99.8% data accuracy supporting procurement and manufacturing operations across corporate and local databases.

★ **RESULTS:** HCL Excellence Award 2024 · 30% obsolescence risk reduction · Zero production stoppages · 20% MTBF improvement · 15% cost savings · 73% faster qualification cycles

ASSISTANT ENGINEER – HARDWARE ENGINEERING & COMPONENT QUALIFICATION

Yazaki India Pvt. Ltd. | Bengaluru, India | November 2018 – June 2022

Hardware design, component qualification, and quality engineering for automotive-grade PCBAs across multi-product automotive electrical systems.

- ◆ **Hardware Architecture & Schematic Design:** Developed hardware specs, electrical schematics, and component datasheets using ORCAD and Altium Designer for ECUs, instrument clusters, and BCMS; ensured DFM/DFT compliance and IATF 16949 adherence.
- ◆ **CAN-Based Instrument Cluster (98% FPY):** Designed automotive-grade instrument cluster using Cypress MCU on CAN bus; led component durability and qualification validation achieving 98% first-pass yield at global launch.
- ◆ **HEV DC-DC Converter | ISO 26262 ASIL-B:** Evaluated 48V→12V power components and led dual-source qualification of IGBTs/MOSFETs — full ASIL-B compliance with zero field failures.
- ◆ **Environmental & EMC Testing (30% Failure Reduction):** Led hardware qualification testing (temperature cycling, vibration, humidity, EMC); implemented CAPA via A3/8D/5WHY reducing field failure rates by 30%.
- ◆ **BOM & Obsolescence Management (8% Cost Reduction):** Managed BOM scrubbing and EOL analysis for 60+ PCBAs and 100+ ECU designs; FFF alternates achieved 8% cost reduction with zero quality compromise.
- ◆ **Kaizen Excellence (45% Defect Reduction):** Led 6+ Kaizen projects across 60+ automotive PCBAs; standardized DFM/DFT protocols delivering 45% defect reduction and 22% cycle-time improvement over 3.5 years.

★ **RESULTS:** 98% FPY at global automotive launch · 45% defect reduction · 30% field failure rate reduction · Manufacturing Excellence Award

PREVIOUS EXPERIENCE

IT Analyst — Component Engineering & Lifecycle Management

Tata Consultancy Services · Pune
Jun 2022 – Nov 2022

Strategic consulting engagement supporting enterprise PLM/component engineering operations during career transition to aerospace hardware roles.

Product Engineer — Power Electronics

Aethon Energy · Bengaluru
Apr 2017 – Oct 2018

Component testing and failure analysis for DSP-Based Sine Wave UPS (1–15 KVA). Achieved 98% FPY, Manufacturing Excellence Award.

Senior Engineer — Product Development & Testing

Solar Cooperation · Bengaluru
Mar 2016 – Mar 2017

Product development, testing, and quality assurance for solar electronics and power management systems.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

Adarsh Management Institute of India, Mahatma Gandhi University | 2015

Diploma in Electronics, Electrical, and Communication Engineering

Nettur Technical Training Foundation | 2011

CERTIFICATIONS & PROFESSIONAL TRAINING

- Cyber Security for Physical Systems and Functional Safety – TUV Saar Certified
- ISO 9001:2015 & ISO 9001:2008 Quality Management Systems Internal Auditor
- ISO 14001:2004 Environmental Management System (EMS) Internal Auditor
- ISO 27001:2013 Information Security Management Internal Auditor – TUV Saar Certified

KEY TECHNICAL PROJECTS

CE Lifecycle Dashboard

HCL Technologies

Integrated SAP + IHS Markit + supplier data into real-time PCBA risk monitoring for 60+ assemblies. Deployed automated PCN/PDN alert workflows reducing manual PLM review by 80%.

✦ Zero production stops · HCL Excellence Award

E6 Global-B Instrument Cluster NPI

HCL Technologies —

Component engineering for 150+ parts (semiconductors, passives, displays, connectors). Environmental stress validation and alternate qualification for global automotive NPI.

✦ 98% FPY · Zero supplier escapes · On-time global launch

DSP-Based Sine Wave UPS Cum-Inverter (1–15KVA)

Aethon Energy

Led component testing and failure analysis during pilot production and mass manufacturing phases, identifying root causes and implementing corrective actions
Collaborated with sourcing teams negotiating pricing, lead times, and quality requirements for power semiconductors and magnetic components—achieving 98% first-pass yield.

✦ Zero production stops · Manufacturing Supporting Excellence Award

HEV DC-DC Converter — ASIL-B

HCL Technologies — ISO 26262

Dual-source qualification of IGBTs/MOSFETs for 48V–12V HEV converter. Full EMC validation, derating analysis, and environmental stress testing for ASIL-B compliance.

✦ ASIL-B compliance achieved · Zero field failures

Kaizen PCBA Risk Mitigation Program

Yazaki India — 60+ PCBA Boards

Led 6+ Kaizen projects standardizing DFM/DFT protocols and structured 8D corrective actions. Sustained quality improvements over 3.5 years across automotive production lines.

✦ 45% defect reduction · 22% cycle time improvement